Two congruences modulo 3 for the series with $A(3x^2) = 2x^2 + A^2 - 2A^3$

A proof of two conjectures in OEIS A389475

Dev D. Goyal

October 2026

Abstract

The sequence A389475 of the On-Line Encyclopedia of Integer Sequences (OEIS) lists the coefficients of the power series $A(x) = x + x^2 + 7x^3 + 17x^4 + \cdots$ that satisfies $A(3x^2) = 2x^2 + A(x)^2 - 2A(x)^3$. Paul D. Hanna conjectured two congruences modulo 3: $A(x)^3 \equiv A(x^3)$ and $A(x)^2 \equiv x^2 + 2A(x^3)$. We prove both. We also prove that the equation has exactly one normalized solution with integer coefficients; this is not immediate, because computing the coefficients from the equation involves a division by 2 at every step. The first congruence is an instance of a classical identity that holds for every power series with integer coefficients. The second follows from the first, because modulo 3 the left side $A(3x^2)$ of the equation vanishes. The results have been checked by computer and proved formally in the Lean proof assistant. A short further argument, not part of the formal verification, shows that modulo 9 the equation determines A completely, as the inverse of an explicit series.

How this paper was produced. This paper reports the result of an AI research effort directed by Dev D. Goyal. ChatGPT (OpenAI), using the GPT-6 model with Pro reasoning and working autonomously from prompts by Dev D. Goyal, found the proof, wrote its Lean 4 formalization, and wrote a first draft of the paper. Dev D. Goyal chose the problems and asked for proofs, formal verification and revisions, but did not supply proof ideas. Claude Opus 5.5, used with Claude Code, then checked the work independently: it rebuilt the formal proofs from source and confirmed that they rely only on Lean's three standard axioms, recomputed the sequence from its definition compared it with the published terms and tested every claim numerically, and read every step of the proofs. It also wrote the present version of this paper. Dev D. Goyal has reviewed the proofs and is responsible for them; this review is not an independent expert validation. The paper has not been peer reviewed. The section "About this paper" at the end gives the details.

1 Introduction

1.1 The sequence

The *On-Line Encyclopedia of Integer Sequences* (OEIS) [3] is a large public database of integer sequences. Many entries also record *conjectures*: patterns that the contributor observed in the data but did not prove.

This paper is about entry A389475 [1], contributed by Paul D. Hanna in October 2025. It defines a power series $A(x) = a(1)\,x + a(2)\,x^2 + a(3)\,x^3 + \cdots$ with integer coefficients and $a(1) = 1$ by the equation

$$A(3x^2) = 2x^2 + A(x)^2 - 2A(x)^3. \tag{1}$$

On the left, $A(3x^2)$ is the series A with x replaced by $3x^2$, so its coefficient of x^{2n} is $3^n a(n)$ (Section 2 explains such substitutions). On the right, $A(x)^2$ and $A(x)^3$ are ordinary powers. Theorem 3.1 shows that there is exactly one such series. Its first coefficients, starting at $n = 1$, are 1, 1, 7, 17, 147, 445, 2199, 9993, 61837, 281417, ... The entry reads, in part:

```
G.f.  A(x) satisfies A(3*x^2) = 2*x^2 + A(x)^2 - 2*A(x)^3.

(4.a) A(x)^3 == A(x^3) (mod 3) (conjecture).

(4.b) A(x)^2 == x^2 + 2*A(x^3) (mod 3) (conjecture).
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— OEIS A389475, Paul D. Hanna, October 2025

Here "G.f." stands for *generating function*, the series whose coefficients are the terms of the sequence.

1.2 The question

The two conjectures compare power series modulo 3, that is, coefficient by coefficient after reducing every coefficient modulo 3. In (4.a), $A(x)^3$ is the cube of A and $A(x^3)$ is A with x replaced by x^3. The two tables compare the two sides of each conjecture. Each row lists the coefficients of x^n, for $n = 0, \ldots, 31$, of the series named on the left, reduced modulo 3. In each table the two rows agree.

n	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$A(x)^3$	0	0	0	1	0	0	1	0	0	1	0	0	2	0	0	0
$A(x^3)$	0	0	0	1	0	0	1	0	0	1	0	0	2	0	0	0

n	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
$A(x)^3$	0	0	1	0	0	0	0	0	0	0	0	1	0	0	2	0
$A(x^3)$	0	0	1	0	0	0	0	0	0	0	0	1	0	0	2	0

n	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$A(x)^2$	0	0	1	2	0	0	2	0	0	2	0	0	1	0	0	0
$x^2 + 2A(x^3)$	0	0	1	2	0	0	2	0	0	2	0	0	1	0	0	0

n	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
$A(x)^2$	0	0	2	0	0	0	0	0	0	0	0	2	0	0	1	0
$x^2 + 2A(x^3)$	0	0	2	0	0	0	0	0	0	0	0	2	0	0	1	0

1.3 What this paper proves

Main result (Theorem 3.2). *Both conjectures are true:*

$$A(x)^3 \equiv A(x^3) \pmod 3 \qquad \text{and} \qquad A(x)^2 \equiv x^2 + 2A(x^3) \pmod 3.$$

We also prove that (1) has exactly one normalized solution with integer coefficients (Theorem 3.1). This needs an argument: when the coefficients are computed from (1) one at a time, each new coefficient $a(n)$ first appears in degree $n + 1$, in $A(x)^2$, with the factor 2, so each step involves a division by 2.

The first congruence holds for every power series with integer coefficients, by a classical identity. For the second, note that every coefficient of $A(3x^2)$ is divisible by 3, so modulo 3 the equation says $0 \equiv 2x^2 + A(x)^2 - 2A(x)^3$; rearranging and using the first congruence gives the second.

1.4 What was known

The entry records the defining equation, some rewritings of it, and the two statements as conjectures. Hanna also contributed the companion entry A389474 [2], whose inverse series satisfies a closely related equation; it is treated in a companion paper with the same method. Statement (4.a) is a direct consequence of a classical identity (Lemma 2.2); we include its proof for completeness and claim nothing new about it. We are not aware of an earlier proof of (4.b) or of the existence of an integer solution.

1.5 Motivation

Equations in which a simple series is substituted into the unknown series, and the result is compared with powers of the unknown, are a natural source of integer sequences; the OEIS contains many of them. Two questions arise for such equations: whether the solution has integer coefficients, and whether its residues follow a pattern. This entry exhibits both. The existence of an integer solution depends on a division by 2 that is always exact, and the congruences follow from the fact that the left side of the equation vanishes modulo 3.

1.6 How to read this paper

Section 2 explains the notions we need: power series, substitution, compositional inverses, and remainders. Section 3 states the results, and Section 4 proves them. Section 5 describes how the results were checked, by computer and by a formal proof in Lean. Section 6 discusses the scope of the results, some stronger statements, and open questions.

2 Background

This section collects the definitions and facts used later. Readers who know formal power series and congruences can skip it and return when needed.

2.1 Formal power series

A *formal power series* is an expression

$$U(x) = u_0 + u_1 x + u_2 x^2 + u_3 x^3 + \cdots ,$$

determined by its sequence of coefficients $u_0, u_1, u_2, \ldots$; see [5, 4]. No number is ever substituted for x, so questions of convergence do not arise. We write $[x^n]\,U = u_n$ for the coefficient of x^n in U; two series are equal when all their coefficients are equal.

Series are added coefficient by coefficient and multiplied like polynomials:

$$[x^n]\,(UV) = u_0 v_n + u_1 v_{n-1} + \cdots + u_n v_0.$$

Each coefficient of a product is a finite sum. For example, $(1-x)(1+x+x^2+x^3+\cdots) = 1$, so $1 + x + x^2 + \cdots$ is the *multiplicative inverse* of $1 - x$, written $\frac{1}{1-x}$. More generally, every series D with constant term 1 has a unique multiplicative inverse $\frac{1}{D}$, whose coefficients are determined one at a time by $D \cdot \frac{1}{D} = 1$ using only addition, subtraction and multiplication. Whenever a quotient of power series appears in this paper, its denominator has constant term 1, so the quotient is again a power series.

A *commutative ring* is a set with an addition, a subtraction and a multiplication that satisfy the usual rules of arithmetic; examples are $\mathbb{Z}$, $\mathbb{Q}$ and the integers modulo m. For a commutative ring R we write $R[[x]]$ for the set of power series with coefficients in R; for instance, writing $\mathbb{F}_2$ for the integers modulo 2, $\mathbb{F}_2[[x]]$ consists of the series with coefficients modulo 2. We write $U(0)$ for the constant term u_0 of U, and $U = O(x^d)$ when the coefficients of $1, x, \ldots, x^{d-1}$ in U are all zero. Thus $U - V = O(x^d)$ means that U and V have the same coefficient of x^n for every $n < d$. A series with constant term 0 and coefficient of x equal to 1, that is, a series of the form $x + u_2x^2 + u_3x^3 + \cdots$, is called *normalized*.

2.2 Substituting one series into another

If $V(x)$ has constant term 0, we can substitute V for x in U:

$$U(V(x)) = u_0 + u_1V(x) + u_2V(x)^2 + u_3V(x)^3 + \cdots .$$

This is called *composition*, also written $U \circ V$. It makes sense because $V(x)^k$ has no terms below x^k, so only the finitely many terms with $k \le n$ affect the coefficient of x^n. For example, if $U = x+x^2$ and $V = x^2$, then $U(V(x)) = x^2+x^4$. Composition is associative: $U \circ (V \circ W) = (U \circ V) \circ W$.

Substituting a series into itself repeatedly gives its *iterates*: $U^{\circ 2}(x) = U(U(x))$, $U^{\circ 3}(x) = U(U(U(x)))$, and so on, with $U^{\circ 0}(x) = x$ and $U^{\circ 1} = U$. They must not be confused with *powers* $U^2 = U \cdot U$, $U^3 = U \cdot U \cdot U$. (The OEIS sometimes writes $A^k(x)$ for the k-th iterate. In this paper the small circle always means an iterate, and a plain exponent always means a power.)

2.3 Undoing a substitution

A normalized series can always be "undone" by another normalized series.

Lemma 2.1 (Compositional inverse)**.** *For every normalized series U there is exactly one normalized series V with $U(V(x)) = x$, and this V also satisfies $V(U(x)) = x$. If U has integer coefficients, so does V.*

We call V the *compositional inverse* of U. The OEIS calls it the *series reversion* of U. For example, the compositional inverse of $x - x^2$ is $C(x) = x + x^2 + 2x^3 + 5x^4 + 14x^5 + \cdots$, whose coefficients are the Catalan numbers. Indeed $C - C^2 = x$, which says exactly that substituting C into $x - x^2$ gives x.

Proof. Write $V = x + v_2x^2 + v_3x^3 + \cdots$ with unknown coefficients. In $U(V) = V + u_2V^2 + u_3V^3 + \cdots$, the coefficient of x^n (for $n \ge 2$) equals v_n plus terms that involve only $v_2, \ldots, v_{n-1}$. Setting it equal to 0 therefore determines v_n from the earlier coefficients, using only additions, subtractions and multiplications. So there is exactly one V with $U \circ V = x$, and it has integer coefficients if U does. Applying the same construction to V gives a normalized W with $V \circ W = x$. Then $U = U \circ (V \circ W) = (U \circ V) \circ W = W$, so $V \circ U = V \circ W = x$ as well. □

2.4 Remainders and congruences

For an integer $m \ge 2$ we write $a \equiv b \pmod m$ ("a is congruent to b modulo m") when $a - b$ is divisible by m, that is, when a and b leave the same remainder after division by m. The remainder of a on division by m is called its *residue* modulo m; it is one of $0, 1, \ldots, m - 1$.

Addition and multiplication are compatible with congruences, so the residues $0, 1, \ldots, m-1$, added and multiplied modulo m, form a ring, written $\mathbb{Z}/m\mathbb{Z}$. When $m = p$ is a prime we also write $\mathbb{F}_p$. In $\mathbb{F}_2 = \{0, 1\}$ we have $1 + 1 = 0$, so addition and subtraction coincide.

For power series with integer coefficients, $U \equiv V \pmod{m}$ means that *every* coefficient of $U - V$ is divisible by m. Equivalently, the two series become equal when every coefficient is replaced by its residue; we call this *reducing* the series modulo m. The key fact is the following.

Reduction modulo m is compatible with all the operations above. To reduce a sum, a product, a substitution, or an inverse, we may reduce the ingredients first and then compute in $\mathbb{Z}/m\mathbb{Z}$. This holds because every coefficient of the result is obtained from finitely many coefficients of the ingredients by adding, subtracting and multiplying, and these operations are compatible with remainders.

Rings. Some statements below hold over every commutative ring, for example $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{Z}/m\mathbb{Z}$. No division is used in them; this matters because in $\mathbb{Z}/m\mathbb{Z}$ division is not always possible (for instance, 2 has no inverse in $\mathbb{Z}/4\mathbb{Z}$).

2.5 The Frobenius identity

Lemma 2.2 (Frobenius identity)**.** *Let p be a prime. For every power series U with coefficients in $\mathbb{F}_p$, we have $U(x)^p = U(x^p)$.*

Proof. In $\mathbb{F}_p$, the binomial theorem simplifies to $(a+b)^p = a^p + b^p$, because every binomial coefficient $\binom{p}{i}$ with $0 < i < p$ is divisible by p. Also $c^p = c$ for every $c \in \mathbb{F}_p$ (Fermat's little theorem). So $\left(\sum_n u_n x^n\right)^p = \sum_n u_n^p x^{pn} = \sum_n u_n x^{pn} = U(x^p)$. The first equality holds for polynomials by repeated use of $(a+b)^p = a^p + b^p$; it holds for power series because each coefficient of $U(x)^p$ involves only finitely many coefficients of U. □

For $p = 2$ this says that squaring a series modulo 2 doubles every exponent: for instance, $(x + x^3)^2 \equiv x^2 + x^6 \pmod{2}$. In particular, the square of any series has, modulo 2, no terms of odd degree.

3 Results

Theorem 3.1 (Exactly one integer solution)**.** *Equation* (1) *has exactly one normalized solution with integer coefficients: the series R_{-1} of Lemma 4.1 below.*

Theorem 3.2 (Hanna's conjectures)**.** *Modulo* 3*,*

$$\textit{(a)}\quad A(x)^3 \equiv A(x^3) \qquad \textit{and} \qquad \textit{(b)}\quad A(x)^2 \equiv x^2 + 2A(x^3).$$

Statement (a) holds for every power series with integer coefficients in place of A.

4 Proofs

The idea in brief. Equation (1) is the case $s = -1$ of the equation $R(3x^2) = 2x^2 + R^2 + 2sR^3$. Lemma 4.1 shows that this equation has exactly one integer solution, by checking that the division by 2 needed at each step is always exact. For the congruences, we reduce modulo

3: then $A(3x^2)$ vanishes, cubing becomes the substitution $x \mapsto x^3$, and the equation can be rearranged into (b).

4.1 An equation with an exact halving step

Both this paper and its companion paper on OEIS A389474 rest on the following lemma, which concerns the equation

$$R(3x^2) = 2x^2 + R(x)^2 + 2s\,R(x)^3, \tag{2}$$

where s is a fixed integer. Here $R(3x^2)$ is R with x replaced by $3x^2$, so its coefficient of x^{2k} is $3^k r_k$, and $R(x)^2$, $R(x)^3$ are ordinary powers.

Why an argument is needed. Suppose we compute the coefficients $r_2, r_3, \ldots$ of a normalized solution $R = x + r_2x^2 + r_3x^3 + \cdots$ one at a time. The coefficient r_n first appears in degree $n+1$, in the square $R(x)^2 = x^2 + \cdots + 2r_nx^{n+1} + \cdots$, with the factor 2. (In $R(x)^3$ it appears only from degree $n+2$ on, and in $R(3x^2)$ only in degree $2n$.) So each step requires a division by 2, and it is not clear that the result is an integer. The proof shows that the number to be divided is always even.

Lemma 4.1 (Exact halving)**.** *For every integer s there is exactly one normalized series R_s with integer coefficients that satisfies* (2)*.*

Proof. For a normalized series U with integer coefficients, define its *error*

$$E(U) = U(x)^2 + 2s\,U(x)^3 - U(3x^2) + 2x^2,$$

so that U solves (2) exactly when $E(U) = 0$.

Step 1: the error is always even. Reduce modulo 2. By Lemma 2.2 with $p = 2$, $U(x)^2 \equiv U(x^2)$. Since $3 \equiv 1$, also $U(3x^2) \equiv U(x^2)$. The terms $2s\,U^3$ and $2x^2$ vanish modulo 2. So $E(U) \equiv 0 \pmod 2$: every coefficient of $E(U)$ is even. Moreover $E(U)$ has no terms of degree 0, 1 or 2: in degree 2 its coefficient is $1 - 3 + 2 = 0$.

Step 2: changing one coefficient. Let V be another normalized integer series with $V - U = c\,x^r + O(x^{r+1})$, where $r \geq 2$. Then

- $V^2 - U^2 = (V-U)(V+U) = 2c\,x^{r+1} + O(x^{r+2})$, because $V + U = 2x + O(x^2)$;
- $V^3 - U^3 = (V-U)(V^2 + VU + U^2) = O(x^{r+2})$, because $V^2 + VU + U^2 = 3x^2 + O(x^3)$;
- $V(3x^2) - U(3x^2) = O(x^{2r})$, which is $O(x^{r+2})$ because $r \geq 2$.

Hence $E(V) - E(U) = 2c\,x^{r+1} + O(x^{r+2})$. The same computation shows that if $V - U = O(x^{r+1})$, then $E(V) - E(U) = O(x^{r+2})$.

Step 3: construction. Start with $U = x$, and suppose that $E(U) = O(x^{r+1})$ for some $r \geq 2$ (this holds for $r = 2$ by Step 1). Let e be the coefficient of x^{r+1} in $E(U)$. By Step 1, e is even, so $e/2$ is an integer. Replace U by $U - (e/2)\,x^r$. By Step 2 the new error is $E(U) - e\,x^{r+1} + O(x^{r+2}) = O(x^{r+2})$, and the coefficients of U below degree r have not changed. Repeating this for $r = 2, 3, 4, \ldots$ fixes the coefficients of U one at a time. Let U_r denote the series obtained after the step for degree r, so that $E(U_r) = O(x^{r+2})$. The resulting series R_s has integer coefficients, and $R_s - U_r = O(x^{r+1})$ for every r. By Step 2, $E(R_s) - E(U_r) = O(x^{r+2})$, so $E(R_s) = O(x^{r+2})$ for every r; that is, $E(R_s) = 0$.

Step 4: uniqueness. Suppose two different normalized solutions first differ in degree r, by $c \neq 0$. Then $r \geq 2$, and by Step 2 their errors differ by $2c \neq 0$ in degree $r+1$. But both errors are 0, a contradiction. □

4.2 Proofs of the results

Proof of Theorem 3.1. Equation (1) is equation (2) with $s = -1$. Apply Lemma 4.1. □

Proof of Theorem 3.2. (a) This is Lemma 2.2 with $p = 3$; it holds for every series with integer coefficients.

(b) The coefficients of $A(3x^2) = \sum_n a(n)\, 3^n x^{2n}$ are all divisible by 3. So, modulo 3, equation (1) says $0 \equiv 2x^2 + A(x)^2 - 2A(x)^3$, that is,

$$A(x)^2 \equiv -2x^2 + 2A(x)^3 \equiv x^2 + 2A(x)^3 \pmod 3,$$

since $-2 \equiv 1$. By (a), $A(x)^3 \equiv A(x^3)$, which gives (b). □

5 Checking the results

A mathematical proof can contain mistakes, so the results were checked in two independent ways.

Numerical check. We solved (1) coefficient by coefficient with exact rational arithmetic, up to degree 89. All coefficients came out as integers, and they reproduce all 23 terms printed in the OEIS entry. Both congruences of Theorem 3.2 hold for every coefficient of degree below 90. The computation was done with the PARI/GP computer algebra system, using code that was written separately from the formal proofs (file `verification/independent_checks.gp` in the accompanying repository [8]). A computation can only test finitely many cases, so it is a check on the definitions and on the indexing, not a proof.

Formal proof in Lean. Lean [6] is a *proof assistant*: a program in which mathematical statements and their proofs are written in a precise formal language, and every step is checked mechanically down to the basic rules of logic. The theorems of Section 3 are proved in Lean 4 (version 4.35.0) on top of Mathlib [7], Lean's library of formalized mathematics. The statements are collected in the file `lean/Entries/A389475.lean` of the repository, in the namespace `OEIS.A389475`. Lemma 4.1 is `integral_family_existence`. The constructions are in `lean/Families/CubicReversion.lean`, and the congruences use `lean/Families/CubicFrobenius.lean` and `lean/Families/CubicConsequences.lean`. The table lists the results covered by the Lean formalization. The Lean formalization does not cover the additional arguments and observations in Section 6.

Statement in this paper	Lean theorem(s)
Lemma 4.1, for every integer s	`integral_family_existence`
Theorem 3.1: A is normalized, solves (1), and is the only normalized integer solution	`normalized`, `source_equation`, `source_unique`, `source_existence_uniqueness`
Theorem 3.2(a), as a series and coefficient by coefficient	`cube_mod_three`, `cube_coefficients`
Theorem 3.2(b), as a series and coefficient by coefficient	`square_mod_three`, `square_coefficients`

The Lean project was rebuilt from source, and the axioms on which each of these theorems depends were listed with Lean's `#print axioms` command. Every one depends only on the three standard axioms of Lean's logic (`propext`, `Classical.choice`, `Quot.sound`);

no step is left unproved. A formal proof certifies exactly the statement that Lean checks. Whether that statement says what the OEIS entry means is a separate question, so the Lean definitions were also compared with the definition in the OEIS entry (Section 1.1). The principal theorems are stated once more, using only Mathlib's definitions, in the file `lean/Challenges/Halving.lean`, and Lean's Comparator tool confirms that the proved theorems have exactly these statements.

6 Discussion

Scope. Theorem 3.2 settles both conjectures of the entry completely. Statement (a) is not special to this sequence.

Modulo 9 the series is determined. The same observation gives more modulo 9. Since $a(1) = 1$ and 9 divides 3^n for $n \geq 2$, we have $A(3x^2) \equiv 3x^2 \pmod 9$, and (1) becomes

$$A(x)^2 - 2A(x)^3 \equiv x^2 \pmod 9.$$

(Modulo 27 one gets in the same way $A(x)^2 - 2A(x)^3 \equiv x^2 + 9x^4$, because $a(2) = 1$.) This determines A modulo 9: *A is congruent modulo 9 to the compositional inverse of $V(x) = x\,(1-2x)^{1/2}$.* Here $(1-2x)^{1/2} = \sum_n \binom{1/2}{n}(-2x)^n$ is the binomial series; its coefficients are fractions whose denominators are powers of 2, and such fractions make sense modulo 9 because 2 has an inverse there ($2 \cdot 5 \equiv 1$). *Proof.* Modulo 9, $V(x)^2 = x^2 - 2x^3$, so $V(A(x))^2 = A(x)^2 - 2A(x)^3 = x^2$. The series $W = V(A(x))/x$ therefore has constant term 1 and $W^2 = 1$. Then $(W-1)(W+1) = 0$, and $W+1$ has the invertible constant term 2, so $W = 1$. Thus $V(A(x)) = x$, and A is the inverse of V (Lemma 2.1). We checked this by computer for every coefficient of degree below 90. It is not part of the Lean formalization.

The family. Equation (2) has an integer solution R_s for every integer s. This entry is the case $s = -1$; the compositional inverse of the series of the companion entry A389474 is the case $s = 1$. For that entry, the companion paper shows that $A(x)^2 \equiv x^2 + 2x^3 \pmod 9$. It would be interesting to understand the other members of the family, and the remainders of the coefficients $a(n)$ themselves modulo higher powers of 3.

About this paper

How the work was done. This paper is one of 28 papers on conjectures from the OEIS that came out of an AI research effort directed by Dev D. Goyal [8]. The roles were as follows.

- *Dev D. Goyal* started the project, chose which problems to pursue, asked for proofs, formal verification and revisions, and decided what to publish. Dev D. Goyal did not supply proof strategies or intermediate arguments.
- *ChatGPT* (OpenAI), using the GPT-6 model with Pro reasoning, worked autonomously from those prompts. It searched the OEIS for suitable conjectures, found the proofs, wrote the Lean 4 formalization, and wrote a first draft of this paper.
- *Claude Opus 5.5*, used with Claude Code, checked the work independently. It rebuilt the Lean project from source and confirmed that the theorems listed in Section 5 rely only on Lean's standard axioms; compared the Lean definitions with the OEIS definition; recomputed the sequence with separately written PARI/GP code, compared it with the

terms published in the OEIS, and tested every claim of the paper on many terms; and read every step of the proofs. It then wrote the present version of the paper, expanding the explanations and, in places, writing out an argument in more detail or by a more elementary route. The theorems of Section 3 and the proof strategies are ChatGPT's. The Discussion section, including the short arguments in it, was written by Claude Opus 5.5; its observations were checked by computer and are not part of the Lean formalization.

Dev D. Goyal has reviewed the proofs and is responsible for this paper; this review is not an independent expert validation.

Review status. This paper has not been peer reviewed, and it has not been submitted to a journal or to the OEIS.

Priority. An OEIS entry that labels a statement as a conjecture shows only that its contributor did not know a proof; it does not show that no proof existed. During the research only limited searches for earlier work were made. No claim of priority is made.

Corrections. Please report any error in the issue tracker of the repository [8], where the Lean sources and the numerical checks are also available.

Acknowledgements

The sequence and the conjectures are due to Paul D. Hanna [1], who also contributed the companion entry A389474 [2]. Statement (4.a) is an instance of the classical Frobenius identity and is not claimed as new.

References

[1] P. D. Hanna, entry A389475 (G.f. $A(x)$ satisfies $A(3x^2) = 2x^2 + A(x)^2 - 2A(x)^3$), *The On-Line Encyclopedia of Integer Sequences*, October 2025. `https://oeis.org/A389475`. Text as of October 2026.

[2] P. D. Hanna, entry A389474 (G.f. $A(x)$ satisfies $A(x^2 + 2x^3 + 2A(x)^2) = 3A(x)^2$), *The On-Line Encyclopedia of Integer Sequences*, October 2025. `https://oeis.org/A389474`. Text as of October 2026.

[3] N. J. A. Sloane and The OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*. `https://oeis.org`.

[4] R. P. Stanley, *Enumerative Combinatorics*, Volume 2, Cambridge University Press, 1999. (Chapter 5 treats formal power series, composition and compositional inverses.)

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